

Running in Cold Weather: Exercise Performance and Cold Injury Risk

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ABSTRACT

Runners compete and train in diverse cold-weather environments that may impact performance and cause cold injuries. This article provides an overview of cold-weather effects on running performance. It also presents the cold-weather injuries likely to occur in runners as well as provides risk management guidance to mitigate susceptibility to cold-weather injuries.

INTRODUCTION

Running in cold weather should not be discontinued because of the cold per se because people generally tolerate outdoor activities and cold exposure without adverse effects. Marathons and ultramarathons are run in cold weather months and locations as diverse as Seattle, Austria, Switzerland, Alaska, and the Arctic. However, exercising in cold weather increases the risk of injury (frostbite and hypothermia), especially in those who are unprepared for changes in weather conditions (10). For the most part, humans rely on avoidance of cold to mitigate cold injury risk, by either wearing protective clothing or taking shelter (7). Athletes will, nevertheless, tolerate even severe cold ($< -18^{\circ}\text{C}$) in pursuit of athletic success. For example, runners completed a marathon in -35°C air with 2 frostbite cases reported (3). This short

article will review how cool to cold temperatures influence exercise performance and risk of cold injury during running.

BIOPHYSICS OF COLD EXPOSURE

Changes in core temperature are caused by either a positive or negative change in heat storage. If the body produces more heat than it dissipates, body tissue storage is positive and deep body temperature rises. Conversely, if heat production is less than that lost to the external environment, then heat storage will be negative and deep body temperature will fall (16). We can present these relationships between production and loss mathematically as follows (16):

$$S = M - E - R - C - K$$

where S = heat storage, M = metabolic heat production, E = evaporation, R = radiation, C = convection, and K = conduction. The equation presented has been simplified to only include heat loss from the heat exchange pathways, E , R , C , and K , although heat gain can also happen through these heat exchange mechanisms (18). Furthermore, external work has been deleted from the equation to simplify the relationship. All units are in $\text{W}\cdot\text{m}^{-2}$. When environmental temperature and water vapor pressure are lower than the skin temperature and vapor pressure of water at the skin, this results in heat loss.

Wetted clothing, caused either by rain or sweat trapped in clothing, can increase the decline in deep body temperature in individuals exercising in cold air, compared to exercise with dry clothing. The reason for these differences lies in the physical properties of air and water; the thermal conductivity of water is 25 times that of air (32). Therefore, wet clothing results in rapid dissipation of the heat, which is delivered to the skin by conduction and convection (blood circulation) from the deeper tissues.

Heat gain and an increase in deep body temperature are also possible during cold-weather exercise because “man in the cold is not necessarily a cold man” (1). This is primarily caused by clothing insulation creating a microenvironment leading to warm skin, increasing the need for evaporative heat loss (5). If metabolic heat production is not balanced by equivalent heat losses, then body temperatures may rise. Because individual differences exist in sweating and exercise efficiency, one standard clothing set should not be imposed, as one person may sweat too much (eventually degrading clothing insulation), whereas another person may not have enough protection (7).

KEY WORDS:

frostbite; hypothermia; nonfreezing cold injury; risk management

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Table 1
Summary of human studies examining the effect of cold exposure on aerobic exercise performance

Reference	Sample size	Exercise mode	Intensity	Environments	Findings/results
Patton and Vogel (24)	8 men	CE 60RPM	75–80% $\dot{V}O_{2\max}$ 30-h at each temperature before exercise	20 and -20°C RH 20–40% AV $0.5 \text{ m}\cdot\text{s}^{-1}$	TTE decreased 38% from 20 to -20°C
Galloway and Maughan (17)	8 men	CE 60–70RPM	70% $\dot{V}O_{2\max}$ to exhaustion	4, 11, 21, 31°C air RH 70% AV $0.7 \text{ m}\cdot\text{s}^{-1}$	$\dot{V}O_2$ was higher at 4 and 11°C vs. 21°C TTE highest at 11°C ($93.5 \pm 6.2 \text{ min}$) with 15% reduction at 4°C
Sandsund et al. (31)	8 men	TM 6% grade	55% $\dot{V}O_{2\max}$ for 10 min 60–95% $\dot{V}O_{2\max}$ $4 \times 5 \text{ min}$ 95% $\dot{V}O_{2\max}$ for 10 min (Salbutamol) 85% $\dot{V}O_{2\max}$ for 5 min (Salbutamol)	-15°C air with/ without Salbutamol $23^{\circ}\text{C}/\text{RH } 52\%$ AV $1.5 \text{ m}\cdot\text{s}^{-1}$	TTE shorter at -15°C compared to 23°C $\dot{V}O_2$ was higher at -15°C Salbutamol increased FEV_1 after exercise
Parkin et al. (23)	8 men	CE 80RPM	70% $\dot{V}O_{2\max}$ until exhaustion	3, 20, 40°C RH $<50\%$	TTE highest at 3°C
Carling et al. (4)	9 men	Professional soccer match	Running speeds 0.0–14.3 km/h 14.4–19.7 km/h $\geq 19.8 \text{ km/h}$	≤ 5 , 6–10, 11– 20, $\geq 21^{\circ}\text{C}$	Shortest distances ran in $\geq 21^{\circ}\text{C}$ ($118.7 \pm 6.9 \text{ m}$) Greater distances covered per minute in $\leq 5^{\circ}\text{C}$ ($9.1 \pm 3.8 \text{ m}$)
Sandsund et al. (30)	9 men	TM 6% grade	60% $\dot{V}O_{2\max}$ for 10 min then 67–91% $\dot{V}O_{2\max}$ $4 \times 5 \text{ min}$ followed by $\dot{V}O_{2\max}$ and TTE to exhaustion	-14 and -9°C , -4 and 1°C , 10 and 20°C AV $5 \text{ m}\cdot\text{s}^{-1}$	TTE highest at -14 & -9°C No significant differences in $\dot{V}O_{2\max}$
Renberg et al. (28)	9 women	TM 6% grade	71% $\dot{V}O_{2\max}$ for 10 min then 76–89% $\dot{V}O_{2\max}$ $4 \times 5 \text{ min}$ followed by $\dot{V}O_{2\max}$ and TTE to exhaustion	-14 and -9°C , -4 and 1°C , 10 and 20°C AV $5 \text{ m}\cdot\text{s}^{-1}$	No difference in TTE among different air temperatures

Table from Castellani and Tipton (5).

AV = air velocity; CE = cycle ergometer; FEV_1 = forced expiratory volume in 1 second; MST = mean skin temperature; RH = relative humidity; TM = treadmill; TTE = time to exhaustion; $\dot{V}O_{2\max}$ = maximum oxygen uptake.

Runners need to experiment with different clothing ensembles in different environmental conditions at different metabolic rates to find the right balance.

AEROBIC EXERCISE PERFORMANCE DURING LABORATORY COLD AIR EXPOSURE

Exercise performance studies in cold air are sparse. For experiments conducted in air, most studies have used a time to exhaustion (TTE) test and compared cold air exposure to either relatively warmer ambient

temperatures (T_{amb}) or those defined typically as temperate environments ($\sim 20^{\circ}\text{C}$). Table 1 lists the studies that have examined aerobic performance during cold air exposure. Overall, there is no consensus about whether exercising in cold air, compared with relatively warmer T_{amb} , impacts aerobic performance. Some studies would suggest that cold T_{amb} impairs TTE (17,24,30,31), whereas others suggest an improvement (4,23) or no change (8,28). In the 2 laboratory studies that systematically used multiple air temperatures, an inverted U pattern

emerged. TTE was maximal at some intermediate ambient temperatures (11°C in shorts/t-shirt; -4 and 1°C with cross-country ski clothing), and TTE decreased as T_{amb} either decreased or increased from this optimal temperature. However, this work does contrast with the field study by Carling et al. (4) who found that running distances in soccer players, who primarily engage in intermittent exercise bouts, were the same whether the air temperatures were in the ranges of $< 5^{\circ}\text{C}$, $6\text{--}10^{\circ}\text{C}$, or $11\text{--}20^{\circ}\text{C}$. Furthermore, the data from Parkin et al. (23)

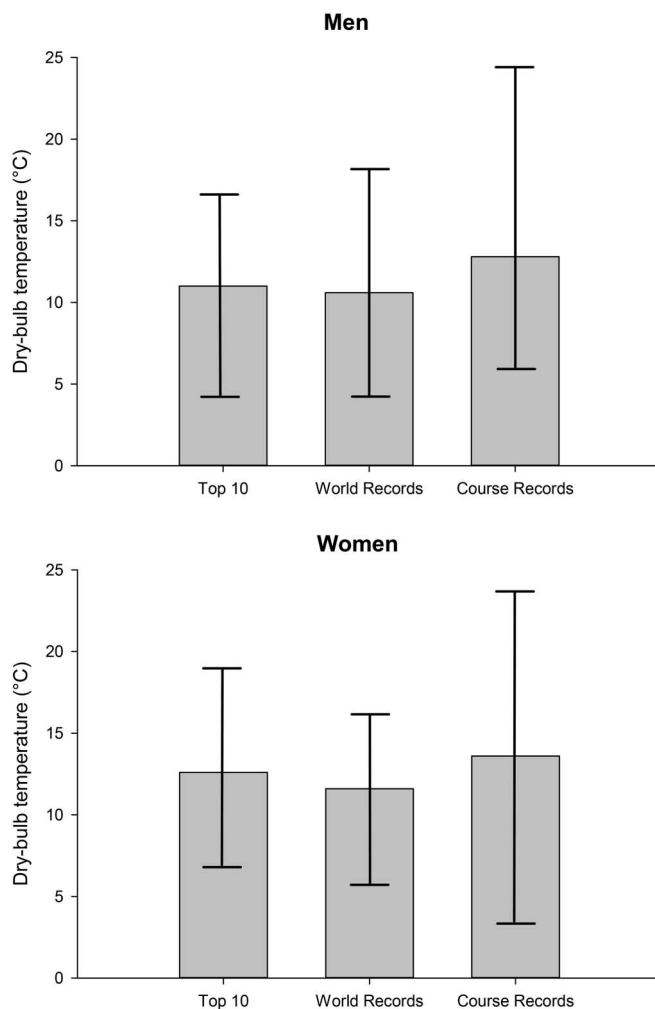


Figure 1. Dry-bulb temperatures during the top 10 fastest ($n = 10$ for men and women), world record (men, $n = 9$; women, $n = 8$), and course records (men, $n = 24$; women, $n = 34$) for the marathon for men and women. Data shown are mean and range. Data are drawn from Ely et al. (15).

directly contrast those observed by Galloway and Maughan (17); they found that TTE while cycling at 70% $\dot{V}O_{2\max}$ was 42% longer in the cold (85 ± 8 minutes) versus temperate (60 ± 11 minutes) environment, whereas Galloway and Maughan found no difference between these 2 air temperatures (4 and 21°C). Cheuvront et al. (8) also did not find any differences between 2 and 20°C air during a 30-minute time trial. Interestingly, the studies above used men as the volunteers. Sandsund and colleagues repeated their study in multiple air environments (-14 – 20°C), but

in women (28), they found no decrement in TTE as T_{amb} fell, in direct contrast to their study in men. They attributed this sex difference to higher heat losses in men, as they exhibited warmer skin temperatures and lower rectal temperatures. Clearly more systematic data are needed to clarify whether cold air impacts long-duration aerobic performance. What is interesting from the different studies is that optimal aerobic performance is likely not to occur during tests conducted in typical laboratory conditions (20–23°C), but in much cooler environments.

Measurement of $\dot{V}O_{2\max}$ has also been used as a proxy for aerobic performance. In the few studies that have used $\dot{V}O_{2\max}$, cold air exposure was found to reduce performance by 5% in -20°C (21,27) and 0°C (27), with comparisons made with 20°C . By contrast, another study found that there were no differences observed in $\dot{V}O_{2\max}$ when comparisons were made between 5 and 18°C air (19). These $\dot{V}O_{2\max}$ results are similar to those reported for studies using TTE as a marker for aerobic performance.

EXERCISE PERFORMANCE IN THE MARATHON

Physical performance is a function of energy output, neuromuscular function, and psychological factors (2). Exercising in cold conditions impacts each of these to potentially reduce performance through changes in cardiovascular function and delivery of fuel, changes in nerve conduction and skeletal muscle contraction, and perception of effort (5). Most of the work has been laboratory-based and used either cycle ergometry or running until exhaustion. However, several articles have examined the effect of ambient temperature (T_{amb}) on running performance, specifically in the marathon.

Most studies have identified that marathon running performance is optimal at ambient temperatures of ~ 8 – 12°C , similar to that observed in the laboratory. Ely et al. (15) identified the top 10 fastest, world record, and course records for the marathon and documented the T_{amb} for these were 11.0, 10.6, and 12.8°C , respectively, for men and 12.6, 11.6, and 13.6°C , respectively, for women. These are shown graphically in Figure 1. Similarly, Vihma (34) showed that 10 – 12°C air temperatures were optimal T_{amb} , whereas Trapasso and Cooper (33) indicated that the best performance in the Boston Marathon was when the wet-bulb globe temperature (WBGT) was below 7.8°C (range of T_{amb} was 2.8 – 21.4°C). Ely et al. (14) used a wet-bulb globe temperature of 5°C as the benchmark temperature to examine the impact of increasing temperature (from 8.6 to 21.9°C) on 42.2-km running

Table 2
Core temperature and associated physiological changes that occur as core temperature falls

Stage	Core temperature		Physiological changes
	°F	°C	
Normothermia	98.6	37.0	
Mild hypothermia	95.0	35.0	Maximal shivering, increased blood pressure
	93.2	34.0	Amnesia, dysarthria, poor judgment, behavior change
	91.4	33.0	Ataxia, apathy
Moderate hypothermia	89.6	32.0	Stupor
	87.8	31.0	Shivering ceases, pupils dilate
	85.2	30.0	Cardiac arrhythmias, decreased cardiac output
	85.2	29.0	Unconsciousness
	82.4	28.0	Ventricular fibrillation likely, hypoventilation
	80.6	27.0	Loss of reflexes and voluntary motion
Severe hypothermia	78.8	26.0	Acid-based disturbances, no response to pain
	77.0	25.0	Reduced cerebral blood flow
	75.2	24.0	Hypotension, bradycardia, pulmonary edema
	73.4	23.0	No corneal reflexes, areflexia
	66.2	19.0	Electroencephalographic silence
	64.4	18.0	Asystole
	59.2	15.2	Lowest infant survival from accidental hypothermia
	56.7	13.7	Lowest adult survival from accidental hypothermia

Table from Castellani et al. (7).

times, again suggesting that marathon performance is optimal at low T_{amb} . El Helou et al. (13) demonstrated that running speeds were optimal in both men and women at $\sim 10^{\circ}\text{C}$. Interestingly, in one marathon (Chicago, 2009) data set analyzed in that study, the T_{amb} was 1.7°C and the statistical fit of the women's data would suggest a reduced running speed at this air temperature compared with 10°C . However, the variability in the data is large, and no firm conclusions can be made regarding a decline in performance at temperatures near freezing, compared with 10°C , and more data are needed.

COLD INJURIES

Cold injuries are classified into 3 categories (11): hypothermia, cold/dry

(freezing), and cold/wet (nonfreezing). Hypothermia is defined as core body cooling below 35°C (95°F). Cold/dry and cold/wet injuries are localized to extremities and exposed skin. Cold/dry injuries cause freezing of cells and tissues and are known as frostbite. Cold/wet injuries are classified as nonfreezing cold injuries (NFCI) and include chilblain and immersion/trench foot. It is not unusual for both hypothermia and local cold injuries to occur simultaneously.

HYPOTHERMIA

With whole-body cooling, hypothermia is clinically defined by a core temperature below 35°C . Initial symptoms include shivering, apathy, and social withdrawal. As core temperature

continues falling below 35°C , confusion, sleepiness, or slurred speech occurs, the so-called “umbles,” that is, grumbles, mumbles, fumbles, and stumbles (11). Table 2 lists the core temperatures and physiological changes typically associated as body core temperature decreases (7).

Hypothermia can be of real concern during races, primarily when the weather conditions are wet. For example, the 2018 Boston Marathon was called the “run to cover” marathon because the race was run in 5.8°C air with high winds and rain throughout the day. Porter (25) describes hypothermia casualties encountered during the 1982 Farnham (United Kingdom) marathon where the T_{amb} was 12°C , with high winds and rain, and the

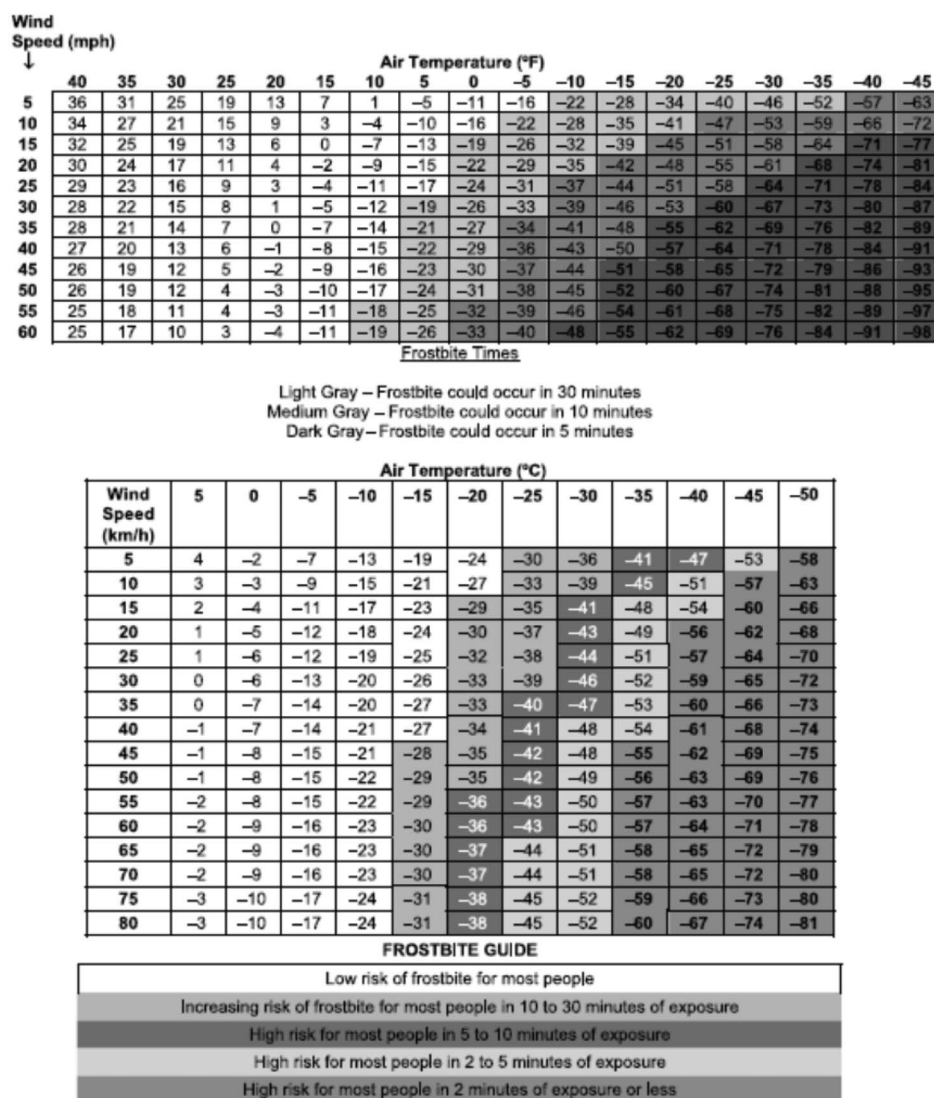


Figure 2. Wind Chill Temperature Index in Fahrenheit and Celsius. Frostbite times are for exposed facial skin. Top chart is from the U.S. National Weather Service; bottom chart is from the Meteorological Society of Canada/Environment Canada.

1984 Guildford race where T_{amb} was 8°C, with winds up to 48 kph (30 mph) and heavy rain. Roberts (29) reported 46 cases of hypothermia over a 12-year span of the Twin Cities (Minneapolis, MN) Marathon. He reported that the highest numbers of nonfinishers (31% of runners) was in 1986, likely due to low wind chill temperatures (-7°C) and sleeting conditions. Another potential issue during long-distance races is the decline in the metabolic rate and heat production when individuals become fatigued. Pugh first suggested that physical exertion might

increase the risk of hypothermia (2). He analyzed reports of 23 separate occasions that had led to numerous cases of hypothermia and 25 deaths. On 1 occasion, although not a running race, 240 men began walking at 06:00 in a drizzle and light wind, vying to be the fastest participants in the Four Inns Walking Competition (3), a combination race/hike traversing 45 miles over the moors of Derbyshire, England (time to complete usually ranged from 9.5 to 22 hours). During this particular contest, the weather became progressively worse with heavy rain and high

winds (25 knots). Only 22 competitors (9.2%) completed the race that year (compared to 66–85% typically observed) and 3 died from hypothermia.

HYPOTHERMIA PREVENTION

Hypothermia can occur at any ambient temperature that enables body heat losses to exceed the metabolic rate. Even exposures to temperatures well above freezing can lead to hypothermia if the metabolic rate is low, and wet/windy conditions facilitate cooling relative to still, dry-air

conditions. If runners can sustain relatively high metabolic rates ($>50\% \dot{V}O_{2\max}$), the risk of hypothermia is lowered. For example, Weller et al. (35) showed that exercise at this threshold maintained core temperature for 4 hours in 4°C air while wearing wet clothing in relatively windy conditions. However, if the exercise intensity is lowered (to $\sim 35\% \dot{V}O_{2\max}$), core temperature falls in cold, wet, and windy conditions (6,26,35). In these conditions, runners should be provided with shelter or extra clothing before and after races. In addition, runners should be careful during training sessions in cold, wet conditions and ensure access to shelters and run with a buddy.

FROSTBITE

Frostbite occurs when tissue temperatures fall below 0°C . The freezing point of skin is slightly below the freezing point of water due to the electrolyte content of the cells and extracellular fluid, so dry tissue typically freezes around -2.2°C (9). Frostbite is most common in exposed skin (nose, ears, cheeks, and exposed wrists) and also occurs in clothed hands and feet because peripheral vasoconstriction can significantly lower tissue temperatures. The most common initial symptom of frostbite is numbness. During rewarming, pain is significant, and rewarming should be clinically supervised for all but the most minor injuries.

FROSTBITE PREVENTION

The Wind Chill Temperature Index (WCTI, Figure 2) is the best indicator of frostbite risk. It takes into account the air temperature, and wind speed and estimates the rate at which tissue cools (20,22). The WCTI, as currently devised, is used to predict the freezing time of exposed facial skin (12). However, the WCTI was developed for individuals who are not exercising, so, in runners who are exercising at high intensities, it will overestimate the risk of frostbite. One important point is that runners will create their own wind; therefore, the speed that individuals are running at needs to be accounted

for when estimating the WCTI. Generally, the risk of frostbite is less than 5% when the ambient temperature is above -15°C , but increased safety surveillance is warranted when the WCTI decreases below -27°C , as frostbite can occur in 30 minutes or less (7).

NONFREEZING COLD INJURIES

The most common NFCIs are trench foot/immersion foot and chilblains. Trench foot typically occurs when tissues are exposed to temperatures between 0 and 15°C for prolonged periods of time (>12 hours), whereas chilblains, a more superficial injury, can occur after just a few hours of exposure of bare skin (11). These injuries may occur due to actual immersion, rain, or by the creation of a damp environment inside boots caused by sweat-soaked socks. The injury classification of NFCI is determined by the symptoms and stages of recovery that occur days and weeks after the exposure. NFCIs would be more likely during ultramarathon races that last for many hours and days.

COLD INJURY RISK MANAGEMENT/ PRACTICAL APPLICATION

Successful management of cold stress requires athletes, coaches, and race officials to appreciate the nature of health hazards associated with running in cold environments and use suitable countermeasures to minimize the risk of hypothermia and frostbite. With proper surveillance and oversight, runners can compete and train safely in most cold-weather environments, and there is usually no need to cancel events. Risk mitigation strategies for runners who are training on extreme cold-weather days include running close to home, taking a mobile phone, and running with a buddy. Runners also need to be cognizant of poor footing from ice/snow that may cause injuries. Runners who are not familiar with running in cold weather should experiment with layering systems and how much they sweat in cold-weather clothing. Runners also need to be aware of changing weather conditions (e.g., possibility of rain) and determine whether they need to bring along

extra clothing if they will be far from shelter. Formal risk management processes have been established to identify potential hazards, contributing factors, and effective controls and countermeasures for employment before/during/after training and competition to prevent cold injuries to athletes. See the American College of Sports Medicine's position stand for more details (7).

SUMMARY

Running in cold weather can be done safely. For long-distance running events such as the marathon, cool weather (8 – 12°C) has been shown to result in the best performance times; similar results have been shown in laboratory studies. Cold injuries are a concern during training and competition. Employment of formal risk management processes such as those recommended by the American College of Sports Medicine (7) can effectively mitigate those risks.

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